Spatial–Temporal Ensemble Convolution for Sequence SAR Target Classification

Ruihang Xue, Xueru Bai, *Member, IEEE*, and Feng Zhou, *Member, IEEE*

***Abstract*— Although numerous methods based on sequence image classification have improved the accuracy of synthetic-aperture radar (SAR) automatic target recognition, most of them only concentrate on the fusion of spatial features of multiple images and fail to fully utilize the temporal-varying features. In order to exploit the spatial and temporal features contained in the SAR image sequence simultaneously, this article proposes a sequence SAR target classification method based on the spatial–temporal ensemble convolutional network (STEC-Net). In the STEC-Net, the dilated 3-D convolution is first applied to extract the spatial–temporal features. Then, the features are gradually integrated hierarchically from local to global and represented as the united tensors. Finally, a compact connection is applied to obtain a lightweight classification network. Compared with the available methods, the STEC-Net achieves a higher accuracy (99.93%) in the moving and stationary target acquisition and recognition (MSTAR) data set and exhibits robustness to depression angle, configuration, and version variants.**

***Index Terms*— Automatic target recognition (ATR), deep learning, image sequence, synthetic-aperture radar (SAR), three-dimensional (3-D) convolution.**

I. Introduction

SYNTHETIC-APERTURE radar (SAR) has played a significant role in military and civilian applications on account of its all-weather, day-and-night, long-range, and high-resolution imaging superiorities [1]. Compared with the optical image, the gray-scale single-polarized SAR image is anisotropic and prone to be interfered by strong jamming or background clutter, which will challenge the available automatic target recognition (ATR) techniques [2].

For some SAR modes, e.g., the spotlight or circular SAR, the imaging radar may observe the same target continuously for a long time and generate chronologically arranged image sequences [3]. The subimages in the sequence usually exhibit conspicuous variations, which contain significant information for classification [4]. Specifically, the structural features of the subimage correspond to spatial features, whereas the variant features of the whole sequence correspond to temporal features. To develop the effective and robust SAR target classification algorithms, it is necessary to fully exploit the characteristics of both the single image and the image sequence. Therefore, how to extract and integrate the spatial–temporal features at different scales has received intensive attention in the SAR ATR community [5]–[13].

A. Prior Work

The target classification task consists of two key issues: feature extractor and classifier [14]. Typical feature extractors, which convert the image into abstract features, include geometric property estimation [15], image processing and transformation [16], [17], principal component analysis (PCA), and linear discriminant analysis [18]. For the design of the classifier, the available methods are mainly based on the minimum mean square error criterion [19], AdaBoost [20], support vector machine [21], and sparse representation [22].

With the development of machine learning, some data-driven classifiers have been proposed, for instance, Wu *et al.* [23] applied a transfer learning to obtain fixed semantic information in each layer of convolutional neural network (CNN) and then adapted it to different data types by altering the weights of middle layers. Such a technique achieves a high accuracy in pol-SAR image classification. Sunaga *et al.* [24] modified the traditional real-valued convolution operation in CNN to complex-valued data to extract the features of the InSAR image. Stamatis and Aouf [25] proposed a multimodal and multidiscipline data fusion strategy appropriate for SAR ATR, which fuses an adaptive sparse coding scheme with a modified clustered AlexNet CNN for final classification. Such a feature extractor–classifier framework effectively avoids hand-designed features that require a large amount of expertise and computational burden and has been successfully applied to single SAR image classification.

For sequence SAR images, the available multiview or multiaspect classification algorithms assume that the images are obtained by the variation of azimuth angles [26], and they fuse the multiple images to obtain a unique category [12]. According to the data fusion level, the available algorithms can be divided into three categories: decision-level fusion, feature-level fusion, and pixel-level fusion.

1) The decision-level fusion algorithms implement feature extraction and classification of each image

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independently and then determine the category through a certain criterion. For instance, the major voting criterion assigns the target type to the majority output [5]. The ranking-based criterion calculates and sorts a normalized score of the target correlation coefficient [6] or classification posterior probability [7] and then takes the maximum expectation of the score as the category. The decision-level fusion achieves an ostensibly improved performance than single image classification; however, it is over-reliant on the target classifier and neglects the correlation among the sequence.

2) The feature-level fusion algorithms combine the features extracted from each image into a vector for classification. Liao *et al*. [8] converted images into a sequence of high range resolution (HRR) profiles, extracted the features of their high-order moments, and then fused them by the hidden Markov model. As for data-driven models, the CNN extracts the image features hierarchically and then concatenates the corresponding feature maps of the sequence image in different network layers [9]. Besides, the long short-term memory (LSTM) is applied to fuse the features of multiple images extracted by the Gabor filter and the three-patch local binary pattern [10]. Although the feature-level fusion algorithms consider the intercorrelation among images, they suffer from structural information loss due to independent feature extraction.
3) The pixel-level fusion algorithms represent the sequence images directly in a unitary feature domain for classification. For instance, Huan and Pan [11] flattened each subimage in the sequence into a vector to calculate the covariance matrix. Then, it constructs the fusion feature map according to the maximal eigenvalue and eigenvector of the covariance matrix and performs the classification based on wavelet PCA. Besides, joint sparse representation (JSR) is also performed in pixel-level fusion, where the target label is selected by the minimal reconstruction error criterion [12], [13].

Although the algorithms discussed above have improved the robustness of SAR ATR by data fusion, they fail to exploit the temporal features contained in the image sequence, i.e., the variation of target scattering characteristics induced by structural occlusion or anisotropy. As the variant temporal features cannot be extracted or identified directly, it is necessary to study the well-designed classification algorithms for sequence models.

The recurrent neural network (RNN), which assigns the sequence data to a hidden layer and propagates it along time steps, has become popular for sequence modeling in recent years [27]. However, the inherent gradient vanishing [28] makes it difficult to train RNN effectively. Later, the LSTM [29] and gated recurrent unit [30] (GRU) modify the connection of time steps in RNN by gates and memory cell, which could improve the propagation mode of the hidden layers and enhance the ability to handle the long-term features. In LSTM, the memory cell stores the historical information of observed sequences. The input gate protects the information stored in the memory cell from perturbation by irrelevant inputs, and the forget gate protects the other units from perturbation by currently irrelevant information stored in the memory cell. The output gate controls how much information of the memory cell is passed to the output layer. Although being widely applied in natural language processing and human behavior recognition, the LSTM and GRU still have difficulties in image sequence processing and parallel training [31].

To deal with the abovementioned problems, scholars have proposed the temporal convolutional network (TCN) for sequence modeling [32]. TCN is a neural network that represents the long-term sequence features by multilayer convolutions. The causal dilated convolution and residual connection of TCN assign the input sequence to the output with the same length, and each element of the output contains the temporal information of the input sequence. Compared with the temporal information in the hidden layers of LSTM, TCN represents the features hierarchically through the predefined convolutional kernels so that the length of the temporal information is flexible and controllable. Meanwhile, the sharable kernels allow TCN to be fed with a batch of sequence data and trained in parallel, which could reduce the heavy computational burden for network training and evaluation.

The sequence model such as LSTM and TCN can only handle one-dimensional (1-D) signals and fails to extract features directly from the two-dimensional (2-D) images [33]. In view of this, some scholars implement two-step methods on optical image sequence, which reduce the dimension of each image by CNN first and then perform the sequence description or action segmentation through LSTM [34] or TCN [32]. During dimension reduction, the image is blurred by convolution or pooling kernels, which will destroy the elusive temporal features among images and cause the loss of sequence information. Therefore, ideal methods for sequence image classification should extract the spatial and temporal features simultaneously and achieve spatial–temporal ensemble representations.

As an extension of CNN, the three-dimensional (3-D) convolution is capable of extracting the spatial–temporal features in a round to form a 3-D tensor. Accordingly, it can handle the higher dimensional data and conduct a synchronous calculation in different dimensions, e.g., in the tasks of video analysis [35] and multispectral crop classification [36]. However, the 3-D convolution fails to represent the long-term sequence information and cannot be treated as a sequence model directly [37]. Besides, it has more learnable parameters than typical CNN, whereas the number of training images is usually limited for SAR targets. Consequently, the common classification strategies for optical images, i.e., to pretrain the generalized network first with a large data set and then fine-tune it [38], may cause serious overfitting for SAR ATR [39].

B. Contribution

To achieve an effective sequence SAR target classification, this article proposes the spatial–temporal ensemble convolutional network (STEC-Net). To fully exploit the spatial and

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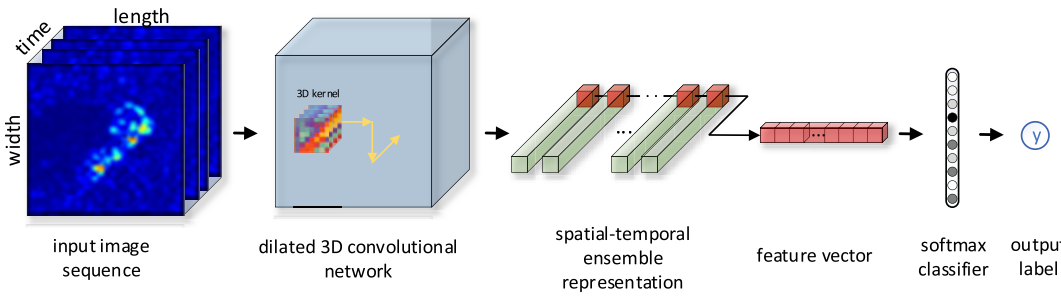


Fig. 1. Overall structure of the STEC-Net.

temporal features, the network introduces causal and dilated calculation into 3-D convolution. Then, it proposes a compact connection to reduce the number of learnable parameters and alleviate overfitting. Compared with the available SAR target classification methods, the main contributions of the STEC-Net include the following.

1) The STEC-Net extracts the spatial and temporal features simultaneously by 3-D convolution and conducts pixel-level fusion at each layer of the network. By treating the spatial and temporal features with the same priority, it effectively avoids the sequence information loss.
2) The STEC-Net represents the temporal features of the sequence as clear and long-term dependence. As the casual dilated convolution has fine-grained control over the length of the temporal memory, each element along the temporal dimension is traceable. Meanwhile, the scale of parameters is adjustable and limited by the sequence length.
3) The overfitting is alleviated in the training phase. With a limited number of parameters, zero padding of the dilated kernel expands the perception fields, and the compact connection simplifies the problem of target label inference. Hence, a lightweight network with streamlined parameters is employed without performance degradation.

The remainder of this article is organized as follows. Section II gives the structure and the principle of the STEC-Net. Section III introduces the training phase of the network, including the forward and backward propagation. Section IV presents the experimental results with detailed analysis. Finally, Section V provides the conclusion and discusses the future work.

II. Structure of the STEC-Net

As shown in Fig. 1, the input image sequence of the STEC-Net can be regarded as a 3-D tensor, where the 3-D corresponds to the sequence time, image length, and image width, respectively. Then, the 3-D kernels of the dilated 3-D convolutional network slide on the 3-D tensor along the length, width, and time dimensions successively to produce the $1 \times 1 \times F \times \mathrm{Ch}$ spatial–temporal ensemble the representation of the image sequence. Specifically, F is the sequence length, and Ch is the channel size of the last layer in the network. As an illustration, we mark the Ch vectors by green bars and denote their last elements by red dice. Such a network structure effectively avoids data decomposition, facilitates the convolution with the 3-D kernels, and simultaneously integrates the data in the sequence time and image pixel dimensions to feature tensors.

For sequence SAR target classification, the three key parts of the STEC-Net are 3-D convolution, dilated convolution, and compact connection. The 3-D convolution conducts hierarchical feature extraction within and among subimages. The dilated convolution covers the large-scale features in the spatial and temporal domains through small kernels. The compact connection pulls the high-dimensional features back to a feature vector to facilitate classification. Further, the three structures will be introduced in detail.

A. 3-D Convolution and 3-D Pooling

Consider an image sequence x with a size of $F \times H \times D$, where F is the number of time steps and $H \times D$ is the size of each subimage; then for typical 2-D multichannel convolution, the output feature map $C_{\text{2-D-mc}}$ for kernel K_{2D} is the summation of all the channels, that is,

$$C_{\text{2-D-mc}}(j,k) = \sum_{i=1}^{F} \sum_{v=0}^{V-1} \sum_{w=0}^{W-1} x(i, j-v, k-w) K_{\text{2-D}}(v,w) \quad (1)$$

where F is the temporal dimension of x, $V \times W$ is the kernel size, $x(i,j,k)$ is the pixel value at (j,k) for the ith subimage, and $K(v,w)$ is the weight of the kernel at (j,k).

As an extension of 2-D convolution in the temporal dimension, the 3-D convolution treats both the input and the output image sequences as 3-D tensors. Specifically, the 3-D convolution is the linear combination of the output in 3-D, that is,

$$C_{\text{3-D}}(i,j,k) = \sum_{u=0}^{U-1} \sum_{v=0}^{V-1} \sum_{w=0}^{W-1} x(i-u, j-v, k-w) K_{\text{3-D}}(u,v,w) \quad (2)$$

where $C_{\text{3-D}}(i,j,k)$ is the value of the feature tensor at (i,j,k), and $K_{\text{3-D}}(\cdot)$ is the 3-D kernel with a size of $U \times V \times W$.

The 2-D multichannel convolution and the 3-D convolution are shown in Fig. 2, where the blue and red patches represent different 2-D kernels or chips of the 3-D kernel. It is noted that each kernel in the 2-D multichannel convolution only generates one feature map [40] regardless of the number of input images. Therefore, the summation in (1) actually

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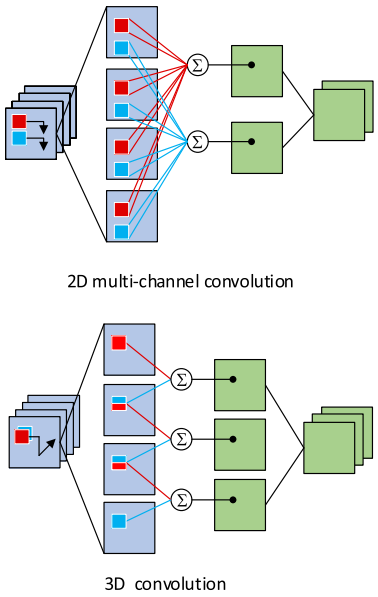


Fig. 2. Comparison between 2-D multichannel convolution and 3-D convolution.

performs an image smoothing along the temporal dimension, which may destroy the temporal features of the sequence [41]. On the contrary, the 3-D convolution connects adjacent images by weights and outputs the 3-D tensors with retained spatial and temporal information. The bottom-up and local-global data fusion of 3-D convolution is natively superior to the decision- and feature-level fusion, which has inherent information loss due to image downsampling.

Usually, the pooling operation follows the convolution to enlarge the receptive fields. Supposing that another 3-D kernel is sliding on the feature tensor A, then the 3-D maximum pooling can be defined as follows:

$$S = \text{maxpool3-D}(A) \tag{3}$$

where the value at (i, j, k) of the pooling output S is calculated by

$$\begin{aligned} S(i,j,k) = \max\{A(i\cdot s_1 + r, j\cdot s_2 + p, k\cdot s_3 + q) \mid \\ r = 0,1,\ldots,R-1; \quad p = 0,1,\ldots,P-1; \\ q = 0,1,\ldots,Q-1\} \end{aligned} \tag{4}$$

where s_1, s_2, and s_3 are the pooling strides, and R, P, and Q are the size of the pooling kernel. If we set $s_1 = 1$, $s_2 = s_3 = 2$, $R = 1$, and $P = Q = 2$, then the output is the maximum value in the $1\times 2\times 2$ cubic of the input feature tensor. It is worthwhile to point out that we set the pooling stride and the kernel length in the temporal dimension to 1 to avoid the dimension reduction. By this means, the loss of sequence memory is prevented.

B. Dilated Convolution

In order to handle a long input sequence, it is feasible to increase the kernel length along the temporal dimension or stack more layers of 3-D convolution, at the expense of computational efficiency and generalization performance.

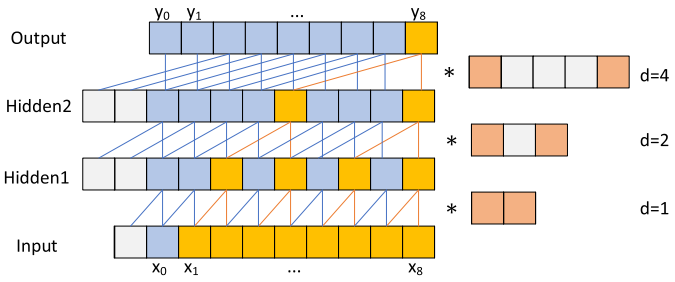


Fig. 3. Illustration of multilayer 1-D dilated convolution.

To solve the contradiction between long-dependence modeling and overfitting, a dilated convolution is proposed [42], which expands the reception field corresponding to the input tensor by padding blanks among kernels. By this means, the number of real kernel parameters will not increase.

For 1-D dilated convolution, the ith element of the output $C_{\text{d1-D}}(i)$ is calculated by

$$C_{\text{d1-D}}(i) = \sum_{u=0}^{k-1} x(i - d\cdot u)\cdot K_{\text{d1-D}}(u) \tag{5}$$

where $K_{\text{d1-D}}(u)$ is the uth element of the kernel, k is the kernel length, and d is the dilation factor. It is noted that a large dilation factor will generate a long-dependence output, and stacking multilayers of dilated convolutions will exponentially increase the information coverage of the last element in the higher layer. Take a 1-D sequence as an example. As shown in Fig. 3, a sequence with a length of 9 is input to the three layers to do a dilated convolution with $d = 1$, 2, and 4, respectively. As indicated by the yellow blocks, the last element of the output y_8 contains information of the eight consecutive elements $x_1 \sim x_8$ from the input although the real kernel size is only 2.

For 3-D dilated convolution, the feature tensor C is obtained by sliding the kernel $K_{\text{d3-D}}$ on the input S with dilation factor $\mathbf{d}$, that is,

$$C = \text{dconv3-D}(\text{S}, K_{\text{d3-D}}, \mathbf{d}) \tag{6}$$

where S, $K_{\text{d3-D}}$, and C are 3-D tensors, the size of $K_{\text{d3-D}}$ is $U\times V\times W$, and $\mathbf{d} = [d_1, d_2, d_3]$. The 3-D dilated convolution is calculated by

$$\begin{aligned} &C(i,j,k) \\ &= \sum_{u=0}^{U-1}\sum_{v=0}^{V-1}\sum_{w=0}^{W-1} S(i - d_1\cdot u, j - d_2\cdot v, k - d_3\cdot w) K_{\text{d3-D}}(u,v,w). \end{aligned} \tag{7}$$

The dilated convolution is performed in each dimension with different physical meanings. In the temporal dimension, the dilated convolution extracts the sequence features. Particularly, the sequence memory passes through layers to break the limitations in kernel size and obtain the representation of the entire sequence. In the spatial dimension, the dilated convolution expands the reception field of the kernel and benefits the maintaining of temporal features in a certain range. When a target is rotating in the image sequence, for example,

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a small reception field only generates the appearing and vanishing of points and lines, whereas an expanded receptive field can capture a continuous variation of the target structure. Besides, the dilated convolution realizes a large receptive field without parameter exploding, which facilitates the training of the STEC-Net.

It should be pointed out that the temporal convolution requires a sequence causality [32], i.e., the current output is only related to the current input and history. In a probabilistic view, the joint distribution of the sequence $x_1, \ldots, x_F$ should satisfy

$$p(x_1, \ldots, x_F) = p(x_1) \prod_{n=2}^{F} p(x_n | x_1, \ldots, x_{n-1}). \tag{8}$$

To guarantee the causality, we pad $(d_1 - 1)(U - 1)$ zero images at the beginning of the temporal dimension for 3-D convolution.

C. Compact Connection

Apart from dilated convolution, the compact connection is also proposed to reduce the number of learnable parameters and alleviate overfitting. In CNN, one or more fully connected layers are designed at the end of the network to blend the extracted features and reduce the dimension of feature maps. However, the fully connected layers have the largest number of parameters and may affect the spatial features. As a substitution, the compact connection replaces the fully connected layer by 3-D convolution. Specifically, for kernels in the last 3-D convolutional layer, their number equals to the number of target labels, and their size equals to the size of the input feature tensor. By this means, the compact connection effectively reduces a large number of parameters in the fully connected layers and generates a feature vector for classification [43], leading to a uniform and concise network.

D. STEC-Net for Sequence SAR Target Classification

The detailed structure of the proposed STEC-Net for sequence SAR target classification is shown in Fig. 4, where the consecutive image sequence with length F is fed into four cascaded 3-D convolution-pooling networks, and the target label is provided by the softmax classifier. Particularly, "dconv3D 128 at $2 \times 4 \times 4$ $d = [1, 3, 3]$" in the first layer means that the 3-D dilated convolution has 128 kernels of size $2 \times 4 \times 4$ and the dilation factor is $[1, 3, 3]$, whereas "maxpool3D $1 \times 2 \times 2$" means that the stride size of the 3-D max-pooling is $1 \times 2 \times 2$. The marks on the other layers have similar meanings. Besides, batch normalization (BN) [44] and a rectified linear unit (ReLU) activation [45] are added behind each layer of the 3-D dilated convolution. Except for the last layer, partial weights of the network are randomly deleted (i.e., dropout) [46] with the probability of 0.1 after each convolutional layer.

The weights in the network are initialized by a truncated normal distribution $\mathcal{N}(0, 0.01)$, whereas the biases are initialized to a constant of 0.1. The learning rate η, which controls the speed of the gradient descent, declines exponentially during training, that is,

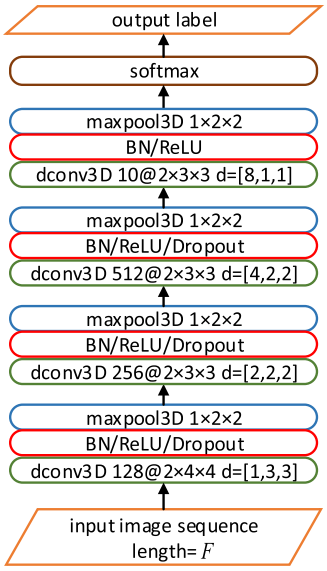


Fig. 4. Detailed structure of the STEC-Net for sequence SAR target classification.

$$\eta = \eta_0 \times d^{\frac{e}{tb}} \tag{9}$$

where η_0 is the initial learning rate, d is the decline factor, e is the index of the current training epoch, and tb is the number of the epochs. In the STEC-Net, we set $\eta_0 = 3 \times 10^{-4}$ and $d = 0.7$. Besides, we set the size of the minibatch to 64.

The dilation factors play different roles in different layers. In the temporal dimension, the dilation factor increases with the layer index, i.e., 1, 2, 4, and 8, whereas the dilation factor decreases in the spatial dimensions, i.e., 3, 2, 2, and 1. As the convolution and pooling operations do not cause downsampling in the temporal dimension, we increase the dilation factors to facilitate information propagation from neighbor to far away in the sequence. In the spatial dimensions, however, the kernels of the higher layer correspond to a large range of the input receptive field due to downsampling by 3-D pooling. Therefore, the dilation factor in a higher layer should be limited to prevent the receptive field from overrunning the image border.

III. Training of the STEC-Net

A. Forward Propagation

According to the typical forward–backward propagation technique [47], the STEC-Net is trained by updating the learnable parameters through gradient descent. Referring to the 3-D convolution defined in (6) and (7), the forward propagation in the lth layer of STEC-Net satisfies

$$\begin{aligned} C_n^{(l)} &= \sum_{m=1}^{M} \text{dconv3-D}\left(S_m^{(l-1)}, K_{nm}^{(l)}, \mathbf{d}^{(l)}\right) + b_n^{(l)} \\ A_n^{(l)} &= f\left(C_n^{(l)}\right) \end{aligned} \tag{10}$$

where $C_n^{(l)}$ is the nth activation of layer l with $l = 1, \ldots, L$, L is the number of network layers, $S_m^{(l-1)}$ is the mth feature

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tensor in layer $l-1$, $K_{nm}^{(l)}$ is the kernel that maps the feature tensors from layer $l-1$ to layer l, and $\mathbf{d}^{(l)}$ and $b_n^{(l)}$ are the dilation factor and the bias parameters, respectively. The output tensor is represented by $A_n^{(l)}$, and $f(\cdot)$ is the nonlinear ReLU activation function, which can be represented as

$$f(\xi) = \max(\xi, 0). \tag{11}$$

Similarly, the forward propagation of the 3-D maximum pooling defined in (3) and (4) is expressed by

$$S_n^{(l)} = \text{maxpool3-D}\big(A_n^{(l-1)}\big) \tag{12}$$

where $S_n^{(l)}$ is the nth pooling output of layer l, and $A_n^{(l-1)}$ is the nth input feature tensor of layer $l-1$.

A compact connection is performed in the fourth convolution layer. Supposing that the length of the image sequence is F and the number of target labels is K, then the output of the compact connection is K tensors $S_k^{(L)}$ with a dimension of $F \times 1 \times 1$, where $k = 1, 2, \ldots, K$. The STEC-Net flattens the tensors to feature the vectors and concatenates their last elements to obtain the output vector O_k. Eventually, the classification probability y_k is given by the softmax classifier defined by

$$\begin{aligned} O_k &= S_k^{(L)}(F-1, 0, 0) \\ y_k &= \text{softmax}(O_k) \end{aligned} \tag{13}$$

where the softmax function is calculated as

$$\text{softmax}(O_k) = \frac{\exp(O_k)}{\sum_{j=1}^{K} \exp\big(O_j\big)}. \tag{14}$$

Finally, the target label y is the index corresponding to the maximum value of y_k, that is,

$$y = \arg\max_k \{y_k\}. \tag{15}$$

B. Backward Propagation

After forward propagation, the STEC-Net compares the inferred output y_k with the real target label t_k and updates the network parameters by optimization. The cross-entropy loss function $E(\mathbf{w})$ is defined as

$$E(\mathbf{w}) = -\sum_{k=1}^{K} t_k \ln y_k(\mathbf{w}) \tag{16}$$

where $\mathbf{w}$ represents all the learnable parameters.

The strategy of optimization is to minimize E by updating $\mathbf{w}$ so that the prediction probability is as close as possible to the target label. As E is a continuous nonlinear function of $\mathbf{w}$, numerical methods can be applied to adjust the network weights along the negative gradient direction, that is,

$$\mathbf{w}^{\text{new}} = \mathbf{w}^{\text{old}} - \eta \nabla E\big(\mathbf{w}^{\text{old}}\big) \tag{17}$$

where $\mathbf{w}^{\text{new}}$ is the updated parameters, $\mathbf{w}^{\text{old}}$ is the original parameters, η is the learning rate, and $\nabla E(\mathbf{w}^{\text{old}})$ is the derivative of E with respect to $\mathbf{w}^{\text{old}}$, i.e., $\partial E / \partial \mathbf{w}^{\text{old}}$.

As the weights are widely distributed throughout the network, we calculate $\nabla E(\mathbf{w}^{\text{old}})$ layer by layer. First, the intermediate values, i.e., the errors, of backward propagation should be obtained. For the output of the network O_k, the error $\delta_k(t)$ is defined by $\partial E / \partial S_k^{(L)}$, which can be deduced by the chain rule as

$$\delta_k(t) = \frac{\partial E}{\partial S_k^{(L)}} = \frac{\partial E}{\partial y_k} \frac{\partial y_k}{\partial O_k} = O_k - t_k. \tag{18}$$

If the $l+1$th layer is 3-D max pooling, then the error δ_n^l is defined as the derivative of the loss function with respect to the activation $A_n^{(l)}$, that is,

$$\delta_n^{(l)} = \frac{\partial E}{\partial A_n^{(l)}} = \frac{\partial E}{\partial S_n^{(l+1)}} \frac{\partial S_n^{(l+1)}}{\partial A_n^{(l)}} \tag{19}$$

where $\partial E / \partial S_n^{(l+1)}$ is the error $\delta_n^{(l+1)}$ propagated by the $l+1$th layer, and $\partial S_n^{(l+1)} / \partial A_n^{(l)}$ is the inverse of the 3-D max pooling, i.e., the 3-D upsampling, which only retains the maximum value in the corresponding pooling window and sets others to 0. Therefore, the error δ_n^l is calculated as

$$\delta_n^{(l)} = \text{upsample}\big(\delta_n^{(l+1)}\big). \tag{20}$$

If the $l+1$th layer is 3-D convolution, then the error $\delta_m^{(l)}$ is defined as the derivative of the loss function with respect to the convolution input tensor $S_m^{(l)}$, that is,

$$\delta_m^{(l)} = \frac{\partial E}{\partial S_m^{(l)}} = \sum_{n=1}^{N} \frac{\partial E}{\partial A_n^{(l+1)}} \frac{\partial A_n^{(l+1)}}{\partial C_n^{(l+1)}} \frac{\partial C_n^{(l+1)}}{\partial S_m^{(l)}} \tag{21}$$

where $\partial E / \partial A_n^{(l+1)}$ is the error $\delta_n^{(l+1)}$ propagated by the $l+1$th layer, $\partial A_n^{(l+1)} / \partial C_n^{(l+1)}$ is the partial derivative of the activation function $f(C_n^{(l+1)})$, and $\partial C_n^{(l+1)} / \partial S_m^{(l)}$ is calculated by the inverse of the 3-D convolution, i.e., the 3-D convolution of $\delta_n^{(l+1)}$ and the kernel rotated by 180°. Therefore, the error δ_m^l can be rewritten as follows:

$$\delta_m^{(l)} = \sum_{n=1}^{N} \frac{\partial f\big(C_n^{(l+1)}\big)}{\partial C_n^{(l+1)}} \text{conv3-D}\big(\delta_n^{(l+1)}, \text{rot180}\big(K_{nm}^{(l+1)}\big), \mathbf{d}^{(l+1)}\big). \tag{22}$$

Similarly, to retain the error dimension after 3-D convolution, we pad $\delta_n^{(l+1)}$ by zeros before further propagation.

IV. Experimental Results

To validate the performance of the proposed STEC-Net for sequence SAR target classification, a series of experiments based on the moving and stationary target acquisition and recognition (MSTAR) data set are performed. Supported by the Defense Advanced Research Projects Agency (DARPA) and the Air Force Research Laboratory (AFRL) [48], the MSTAR data set contains high-resolution spotlight SAR images of ten different ground military vehicles. The X-band imaging radar works in the HH polarization mode, and the obtained image size is 128×128 pixels with a resolution of 0.3×0.3 m. The depression angles of SAR are 15°, 17°, and 30°, and the azimuth angle for each target varies continuously from 0° to 360° with an interval of 5° or 6°.

The original images in the MSTAR data set need to be cropped and arrayed to generate the sequences suitable for classification. Besides experiments under standard operating conditions (SOCs), various extended operating conditions (EOCs) are considered to test the robustness of the

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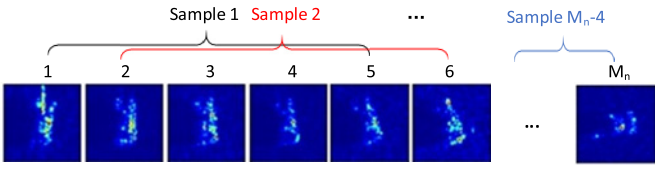


Fig. 5. Generation of the sequence samples.

STEC-Net, such as large depression angle (EOC1), configuration variants (EOC2/C), and version variants (EOC2/V). Then, the STEC-Net is compared with several state-of-the-art algorithms on the original and clutter-removed data set. In order to illustrate the spatial–temporal features extracted by the STEC-Net, we show the ensemble feature tensors in different layers and visualize the high-dimensional tensors as 2-D point sets.

The proposed STEC-Net is implemented on an NVIDIA GTX1080Ti GPU with the framework of TensorFlow.

A. Data Preparation

For the MSTAR data set, where images are continuously arranged in azimuth, we crop the 60×60 rectangular subimage from the center of the original image to retain the target information and reduce the influence of background on network training [9], [39]. Then, we generate the sequence data with a predetermined length F. Taking $F = 5$ as an example, a sampling window slides along the images with a stride of 1 and 5 continuous images in the window are combined as a sample, as shown in Fig. 5. As the images of different targets cannot appear in the same sequence, some cross-category samples are deleted. Therefore, the number of sequence samples is smaller than the number of original images.

The SOC experiment is performed on the data set with ten types of targets, including tank T-62 and T-72, armored carrier BMP-2, BTR-70, BTR-60, and BRDM-2, self-propelled artillery 2S1, bulldozer D7, truck ZIL-131, and antiaircraft gun ZSU-234. The depression angles of the training set and the test set are 17° and 15°, respectively. Table I shows the detailed settings of the training and test sets in the SOC experiment.

To verify the satisfying generalization performance and the robustness of the STEC-Net, the test samples differ from the training samples in EOC scenarios. Specifically, the EOC experiments are divided into three parts, i.e., EOC1, EOC2/C, and EOC2/V.

The EOC1 experiment is carried on four targets (i.e., 2S1, BRDM-2, T-72, and ZSU-234). The depression angles of the training and test sets are 17° and 30°, respectively, which determine the imaging plane and affect the target shape in the SAR image. The detailed settings of the samples are shown in Table II.

The EOC2/C corresponds to configuration variation, where the targets in the training and test sets differ in components such as reactive armor and auxiliary fuel barrel. There are four targets (i.e., BMP-2, BRDM-2, BTR-70, and T-72) in the training set of the EOC2/C experiment, and only, the T-72 target with five configuration variations is in the test set. The EOC2/V corresponds to target version variation, where some structures have a relative motion among the training and test sets, such as the rotation of the tank turret and guns and the opening and closing of the hatch. The training set of the EOC2/V experiment is the same as that of the EOC2/C experiment, but there are seven types of samples in the test set, including five versions of T-72 and two versions of BMP-2. The detailed settings of the samples are shown in Tables III and IV.

TABLE I
TRAINING AND TEST SAMPLES FOR SOC EXPERIMENTAL SETUP

Label	Target Type	Sample Number					
		Train (Depression 17°)			Test (Depression 15°)		
		F=5	F=10	F=15	F=5	F=10	F=15
0	BMP-2	694	689	684	583	578	573
1	BTR-70	229	224	219	192	187	182
2	T-72	687	682	677	578	573	568
3	BTR-60	252	247	242	191	186	181
4	2S1	295	290	285	270	265	260
5	BRDM-2	294	289	284	270	265	260
6	D-7	295	290	285	270	265	260
7	T-62	295	290	285	269	264	259
8	ZIL-131	295	290	285	270	265	260
9	ZSU-234	295	290	285	270	265	260

TABLE II
TRAINING AND TEST SAMPLES FOR EOC1 (LARGE DEPRESSION VARIATION)

Train (Depression 17°)				Test (Depression 30°)			
Type	Sample Number			Type\ Series	Sample Number		
	F=5	F=10	F=15		F=5	F=10	F=15
2S1	295	290	285	2S1\b01	284	279	274
BRDM-2	294	289	284	BRDM-2\E71	283	278	273
T-72	687	682	677	T-72\A64	283	278	274
ZSU-234	295	290	285	ZSU-234\d08	284	279	274

TABLE III
TRAINING AND TEST SAMPLES FOR EOC2/C (CONFIGURATION VARIATION)

Train (Depression 17°)				Test (Depression 15°,17°)			
Type/Series	Sample Number			Type/Series	Sample Number		
	F=5	F=10	F=15		F=5	F=10	F=15
BMP-2/9563	229	224	219	T-72\S7	415	410	405
BRDM-2/E71	229	224	219	T-72\A32	568	553	558
BTR-70/c71	228	223	218	T-72\A62	569	564	559
T-72/132	294	289	284	T-72\A63	569	564	559
				T-72\A64	569	564	559

B. Results of SOC and Feature Visualization

Table V shows the confusion matrix of the SOC experiment when the sequence length F is 5, where each row denotes the

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TABLE IV
TRAINING AND TEST SAMPLES FOR EOC2/V (VERSION VARIATION)

Train (Depression 17°)				Test (Depression 15°,17°)			
Type/Series	Sample Number			Type/Series	Sample Number		
	F=5	F=10	F=15		F=5	F=10	F=15
BMP-2/9563	229	224	219	T-72\812	422	417	412
BRDM-2/E71	229	224	219	T-72\A04	569	564	559
BTR-70/c71	228	223	218	T-72\A05	569	564	559
T-72/132	294	289	284	T-72\A07	569	564	559
				T-72\A10	563	558	553
				BMP-2\9566	424	419	414
				BMP-2/C21	425	420	415

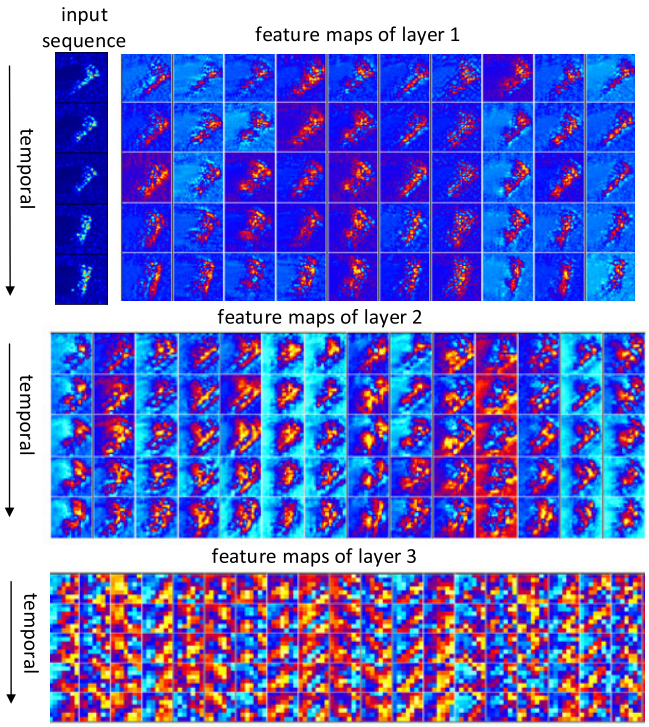


Fig. 6. Input images and feature maps of different layers.

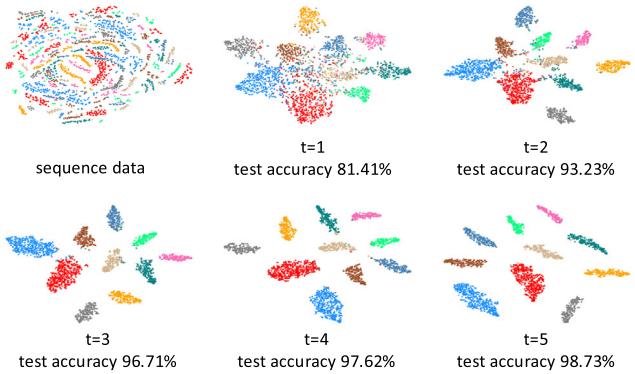


Fig. 7. Visualization of sequence data and time slices.

true target label and each column denotes the predicted class label. It can be observed that except for 2S1 and BRDM-2, the classification accuracy of each target is 100%, and the overall accuracy is 99.53%. Target 2S1 contains a large body and smooth outlines like armored carriers, whereas it has a turret and tracks like tanks, so 2S1 may be misclassified.

Tables VI and VII show the confusion matrices of the SOC experiments for $F = 10$ and $F = 15$, where the classification accuracy is 99.61% and 99.93%, respectively. Adding more images into a sequence sample will further improve the performance of the STEC-Net. For example, the similarity between 2S1 and BTR-70 for $F = 5$ may lead to misclassification, whereas the confusion is eliminated when more features are integrated for $F = 15$.

The STEC-Net is able to achieve a high accuracy for sequence SAR target classification, thanks to its capability of spatial–temporal ensemble feature extraction. As an illustration, the ensemble features are shown in Fig. 6 for the samples of the SOC experiment when $F = 5$. An image sequence of target BMP-2 is fed into the STEC-Net, and the ensemble feature tensors of the first three layers are unfolded and arrayed to feature maps along the temporal dimension. In the feature maps, the submaps of each row denote the features extracted by different kernels, whereas the submaps of each column denote the feature variations along the temporal dimension.

The feature maps of layer 1 are specific, such as the rotation and deformation of the target scattering centers. Based on layer 1, layer 2 extracts the shape and structural features of the target and transits them to target deformation along the temporal dimension. The feature maps of layer 3 are more abstract, and the spatial and temporal information have a closer relationship. Along the temporal dimension, there are changes of displacement, rotation, expansion, and collapse, in the highlighted areas, which indicate that the kernels of layer 3 have decomposed the sequence features to motion tuples.

Fig. 6 shows the feature extraction of the STEC-Net from a microscopic vision. Then, to illustrate the sequential memory of the STEC-Net, Fig. 7 shows a macroscopic vision on the entire test set. For samples of the SOC experiment when $F = 5$, the input test data and the tensors in the last layer are mapped to a 2-D distribution, respectively, by the t-distributed stochastic neighbor embedding (t-SNE) [49]. The t-SNE is a dimension reduction method, which represents the similarity of the high-dimensional data as the conditional probabilities and generates a similar distribution in the low-dimensional space. When the two distributions are almost identical, the projection of the abstract high-dimensional data can be observed in the low-dimensional space.

In Fig. 7, t represents the index of time slices in the feature tensor obtained by compact connection. The points sharing the same color are the projection of the data belonging to the same target. Besides, it provides the classification accuracy corresponding to the five slices. It can be observed that the interclass distance of the features is increasing with the accumulation of slices, which makes it easier for the classifier to distinguish. Since the dilated 3-D convolution has causality, the last time slice accumulates all the sequential information and contributes to a better classification performance.

The comparison between the compact connection and traditional fully connected layer in the STEC-Net is shown

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TABLE V
CONFUSION MATRIX FOR SOC EXPERIMENTAL SETUP ($F = 5$)

Sample Number		Output Label 0	1	2	3	4	5	6	7	8	9	Total	Acc. (%)
Real label	0	583	0	0	0	0	0	0	0	0	0	583	100.00
	1	0	192	0	0	0	0	0	0	0	0	192	100.00
	2	0	0	578	0	0	0	0	0	0	0	578	100.00
	3	0	0	0	191	0	0	0	0	0	0	191	100.00
	4	1	8	0	1	256	3	0	1	0	0	270	94.81
	5	0	0	0	1	0	269	0	0	0	0	270	99.63
	6	0	0	0	0	0	0	270	0	0	0	270	100.00
	7	0	0	0	0	0	0	0	269	0	0	269	100.00
	8	0	0	0	0	0	0	0	0	270	0	270	100.00
	9	0	0	0	0	0	0	0	0	0	270	270	100.00
Total		584	200	578	193	256	272	270	270	270	270	3163	99.53

TABLE VI
CONFUSION MATRIX FOR SOC EXPERIMENTAL SETUP ($F = 10$)

Sample Number		Output Label 0	1	2	3	4	5	6	7	8	9	Total	Acc. (%)
Real label	0	577	0	0	0	0	1	0	0	0	0	578	99.83
	1	0	187	0	0	0	0	0	0	0	0	187	100.00
	2	0	0	573	0	0	0	0	0	0	0	573	100.00
	3	0	0	0	186	0	0	0	0	0	0	186	100.00
	4	0	10	0	0	255	0	0	0	0	0	265	96.23
	5	0	0	0	0	0	264	0	0	1	0	265	99.62
	6	0	0	0	0	0	0	265	0	0	0	265	100.00
	7	0	0	0	0	0	0	0	264	0	0	264	100.00
	8	0	0	0	0	0	0	0	0	265	0	265	100.00
	9	0	0	0	0	0	0	0	0	0	265	265	100.00
Total		577	197	573	186	255	265	265	264	266	265	3113	99.61

TABLE VII
CONFUSION MATRIX FOR SOC EXPERIMENTAL SETUP ($F = 15$)

Sample Number		Output Label 0	1	2	3	4	5	6	7	8	9	Total	Acc. (%)
Real label	0	573	0	0	0	0	0	0	0	0	0	573	100.00
	1	0	182	0	0	0	0	0	0	0	0	182	100.00
	2	0	0	566	0	1	0	0	1	0	0	568	99.65
	3	0	0	0	181	0	0	0	0	0	0	181	100.00
	4	0	0	0	0	260	0	0	0	0	0	260	100.00
	5	0	0	0	0	0	260	0	0	0	0	260	100.00
	6	0	0	0	0	0	0	260	0	0	0	260	100.00
	7	0	0	0	0	0	0	0	259	0	0	259	100.00
	8	0	0	0	0	0	0	0	0	260	0	260	100.00
	9	0	0	0	0	0	0	0	0	0	260	260	100.00
Total		573	182	566	181	261	260	260	260	260	260	3063	99.93

in Fig. 8, where the dimension of the fully connected layer is 1024. It can be observed that the compact connection leads to smaller inner class distance, which is beneficial to the construction of a simple and efficient classifier.

C. Results of EOC

The confusion matrices of the EOC1 experiment for $F = 5$, 10, and 15 are shown in Table VIII, and the corresponding classification accuracy is 96.92%, 98.21%, and 98.53%, respectively. The confusion matrix of the EOC2/C experiment is shown in Table IX, and the accuracy of the five configuration variations of target T-72 is up to 99.81%. The confusion matrix of the EOC2/V experiment is shown in Table X, and the accuracy of the version variations of target BMP-2 and T-72 is up to 99.13%.

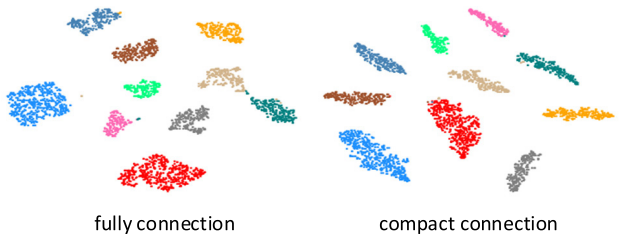


Fig. 8. Visualization of the fully connected layer and the compact connection.

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TABLE VIII
CONFUSION MATRIX FOR EOC1 (LARGE DEPRESSION VARIATION)

Sequence length	Output label	2S1	BRDM	T72	ZSU234	Total	Acc. (%)
F=5	2S1	280	1	3	0	284	98.59
	BRDM-2	0	283	0	0	283	100.00
	T72	2	0	267	15	284	94.01
	ZSU234	2	3	9	270	284	95.07
	Total					1135	96.92
F=10	2S1	273	5	1	0	279	97.85
	BRDM-2	0	278	0	0	278	100.00
	T72	2	16	260	1	279	93.19
	ZSU234	0	2	3	274	279	98.21
	Total					1115	98.21
F=15	2S1	274	0	0	0	274	100.00
	BRDM-2	0	273	0	0	273	100.00
	T72	1	6	258	9	274	94.16
	ZSU234	0	0	0	274	274	100.00
	Total					1095	98.53

TABLE IX
CONFUSION MATRIX FOR EOV2/C (CONFIGURATION VARIATION)

Sequence length	Type/ Series	BMP-2	BTR-70	T-72	BRDM-2	Total	Acc. (%)
F=5	T-72/S7	1	0	414	0	415	99.76
	T-72/A32	2	0	566	0	568	99.65
	T-72/A62	0	0	569	0	569	100.00
	T-72/A63	10	0	559	0	569	98.24
	T-72/A64	26	0	543	0	569	95.43
	Total					2690	98.55
F=10	T-72/S7	1	0	409	0	410	99.76
	T-72/A32	0	0	563	0	563	100.00
	T-72/A62	0	0	564	0	564	100.00
	T-72/A63	0	0	564	0	564	100.00
	T-72/A64	2	0	562	2	566	99.29
	Total					2667	99.81
F=15	T-72/S7	0	0	405	0	405	100.00
	T-72/A32	0	0	558	0	558	100.00
	T-72/A62	0	0	559	0	559	100.00
	T-72/A63	0	0	559	0	559	100.00
	T-72/A64	1	4	554	0	559	99.11
	Total					2640	99.81

TABLE X
CONFUSION MATRIX FOR EOC-2/V (VERSION VARIATION)

Sequence length	Type/ Series	BMP-2	BTR-70	T-72	BRDM-2	Total	Acc. (%)
F=5	BMP-2/9566	401	0	23	0	424	94.58
	BMP-2/C21	421	0	4	0	425	99.06
	T-72/812	2	0	419	1	422	99.29
	T-72/A04	44	0	525	0	569	92.27
	T-72/A05	0	0	569	0	569	100.00
	T-72/A07	21	0	548	0	569	96.31
	T-72/A10	1	0	562	0	563	99.82
	Total					3541	98.10
F=10	BMP-2/9566	411	0	8	0	419	98.09
	BMP-2/C21	420	0	0	0	420	100.00
	T-72/812	1	0	416	0	417	99.76
	T-72/A04	18	0	543	3	564	96.28
	T-72/A05	0	0	564	0	564	100.00
	T-72/A07	4	0	560	0	564	99.29
	T-72/A10	0	0	558	0	558	100.00
	Total					3506	99.03
F=15	BMP-2/9566	402	3	7	2	414	97.10
	BMP-2/C21	412	0	3	0	415	99.28
	T-72/812	6	0	406	0	412	98.54
	T-72/A04	4	1	552	2	559	98.75
	T-72/A05	0	0	559	0	559	100.00
	T-72/A07	2	0	557	0	559	99.64
	T-72/A10	0	0	533	0	533	100.00
	Total					3451	99.13

TABLE XI
PERFORMANCE COMPARISON BETWEEN STEC-NET AND OTHER METHODS

Model	SOC (%)	EOC1 (%)	EOC2/C (%)	EOC2/V (%)
Decision Fusion [7]	96.10	-	-	-
SRC [50]	99.12	-	96.78	-
JSR [12]	94.69	96.50	-	-
A-ConvNet [39]	99.13	96.12	98.93	98.60
MA-BLSTM [10]	99.90	-	99.59	-
VD-CNN [9]	98.52	94.61	95.45	95.46
STEC-Net (*F*=5)	99.53	96.91	98.55	98.10
STEC-Net (*F*=10)	99.61	97.30	99.81	99.03
STEC-Net (*F*=15)	**99.93**	**98.53**	**99.81**	**99.13**

D. Algorithm Comparison

The comparison of the classification performance between the STEC-Net and several available algorithms is shown in Table XI, involving three traditional pattern recognition algorithms and three deep learning algorithms. Among them, all the methods except for the A-ConvNet [39] (which cannot be adjusted for sequence image classification) are carried on the image sequence. For the decision method [7], the classification accuracy is 96.10% for sequence data, which is 4.31% higher than that in the single image. The SRC method [50] extracts the confidence of the sequence images by constructing a global dictionary and then conducts JSR for feature fusion. For the input image group with ten subimages, its classification accuracy is 99.12%. The JSR method [12] applies JSR at the pixel-level fusion, and the classification accuracy is 94.69% for the image group with three subimages. The MA-BLSTM [10] conducts bidirectional LSTM on the image sequence for feature-level fusion, and the classification accuracy is 99.90%. The VD-CNN [9] fuses the features of the input image group with four subimages by a multilevel CNN, and the classification accuracy is 98.52%. The above methods adopt the optimal length of the input image sequence, which could generate the highest classification accuracy with an acceptable computational burden.

As shown in Table XI, the STEC-Net has a higher accuracy for longer image sequences, and the accuracy is higher than those of the available methods, especially in the

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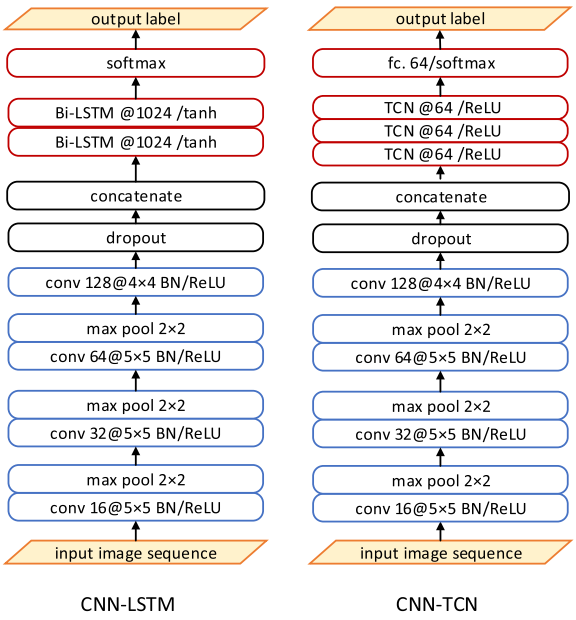


Fig. 9. Network structures for CNN-LSTM and CNN-TCN.

EOC scenarios. Therefore, the STEC-Net has the ability to obtain the robust and generalized classification results for target variants.

E. Results on Clutter-Removed Data Set

For the original MSTAR data set, there is usually a high correlation between clutter and target in the training and test samples [51], which may result in better performance than what would be achieved in the operational scenarios. Therefore, we remove the clutter in SAR images by threshold segmentation [14] and perform experiments under the same conditions as SOC and EOC. In order to compare the accuracy and time consumption of different algorithms, the A-ConvNet in [39] and the VD-CNN in [9] are reimplemented.

Recently, sequence modeling methods based on RNN-like structures have achieved remarkable results in the field of sequence image processing [32], [34]–[37]. However, most of them are dedicated to optical image classification or sequence-to-sequence missions, which cannot be directly applied to SAR ATR. To compare the performance of the proposed STEC-Net with the available methods, we adapt CNN-LSTM [34], CNN-TCN [32], 3-D CNN [36], and T3D Conv. [37] to SAR sequence image classification, respectively. The CNN-LSTM and CNN-TCN are the feature-level fusion methods which apply CNN to extract spatial features and connect them to a sequence. As shown in Fig. 9, the CNN-LSTM processes the feature sequence through a double-layer bidirectional LSTM (Bi-LSTM), whereas CNN-TCN accomplishes this task through a three-layer TCN block. For traditional 3-D CNN, we replace the casual dilated convolution in Fig. 4 by 3-D convolution. Meanwhile, we add a dense connection between the 3-D convolution layers and obtain the T3D Conv.

The classification accuracy and the corresponding training time for a batch in SOC are shown in Table XII. It is observed

TABLE XII
PERFORMANCE COMPARISON WITH CLUTTER-REMOVED IMAGES

Model	Accuracy				SOC batch time (s)
	SOC (%)	EOC1 (%)	EOC2/C (%)	EOC2/V (%)	
A-ConvNet	95.04	89.05	87.31	87.08	**0.12**
VD-CNN	98.29	90.16	93.73	93.63	0.30
CNN-LSTM (F= 5)	97.82	93.30	90.84	95.77	22.21
CNN-LSTM (F=10)	98.45	97.22	93.00	97.04	42.52
CNN-LSTM (F=15)	99.80	99.18	99.13	97.52	69.37
CNN-TCN (F= 5)	98.01	90.84	87.88	86.92	2.94
CNN-TCN (F=10)	99.26	93.00	90.39	87.96	4.65
CNN-TCN (F=15)	99.80	97.17	89.40	91.19	6.73
3D CNN (F=5)	98.99	93.92	92.16	92.77	2.58
3D CNN (F=10)	98.78	95.43	93.80	94.41	4.53
3D CNN (F=15)	99.02	95.80	94.89	93.73	7.77
T3D Conv (F=5)	98.20	90.66	97.77	95.45	2.11
T3D Conv (F=10)	97.91	93.63	96.39	97.69	3.93
T3D Conv (F=15)	98.11	96.43	98.71	96.23	5.63
STEC-Net (F= 5)	98.74	95.77	98.10	97.37	1.78
STEC-Net (F=10)	99.81	97.04	99.10	97.98	3.20
STEC-Net (F=15)	**99.87**	**99.72**	**99.21**	**99.77**	4.25

that after clutter removal, the accuracy of A-ConvNet and VD-CNN decreases, especially in the EOC experiments. Although the CNN-LSTM is competent in handling sequence image, the training time is much longer than the other methods. The CNN-TCN and 3-D CNN require less training time while the classification accuracy drops obviously in the EOC experiments. The accuracy of T3D Conv. is higher than that of the 3-D CNN because of the temporal layer and the dense connection. However, the dense connection between layers leads to an excessive number of parameters, which may lead to overfitting with a limited number of SAR target samples. On the contrary, the proposed STEC-Net achieves a good balance between classification accuracy and training time, providing an effective and efficient solution to sequence SAR image classification.

V. CONCLUSION

To fully exploit the characteristics of continuous SAR imaging, an efficient spatial–temporal ensemble convolutional network, namely, the STEC-Net, was proposed. In the network, the 3-D convolution integrated the information of spatial and temporal features, the dilated convolution realized the long-term representation of sequence images, and the compact connection contributed to a lightweight parameter set. The experiments on the MSTAR data set have demonstrated the effectiveness of the STEC-Net in sequence feature extraction and representation. Compared with the available algorithms, the proposed network has shown a higher classification accuracy with acceptable time consumption. Besides, the STEC-Net has demonstrated robust classification performance on the clutter-removed data set.

The future work will be focused on sequence SAR target classification in complex scenarios, such as small data sets and deformed samples.

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